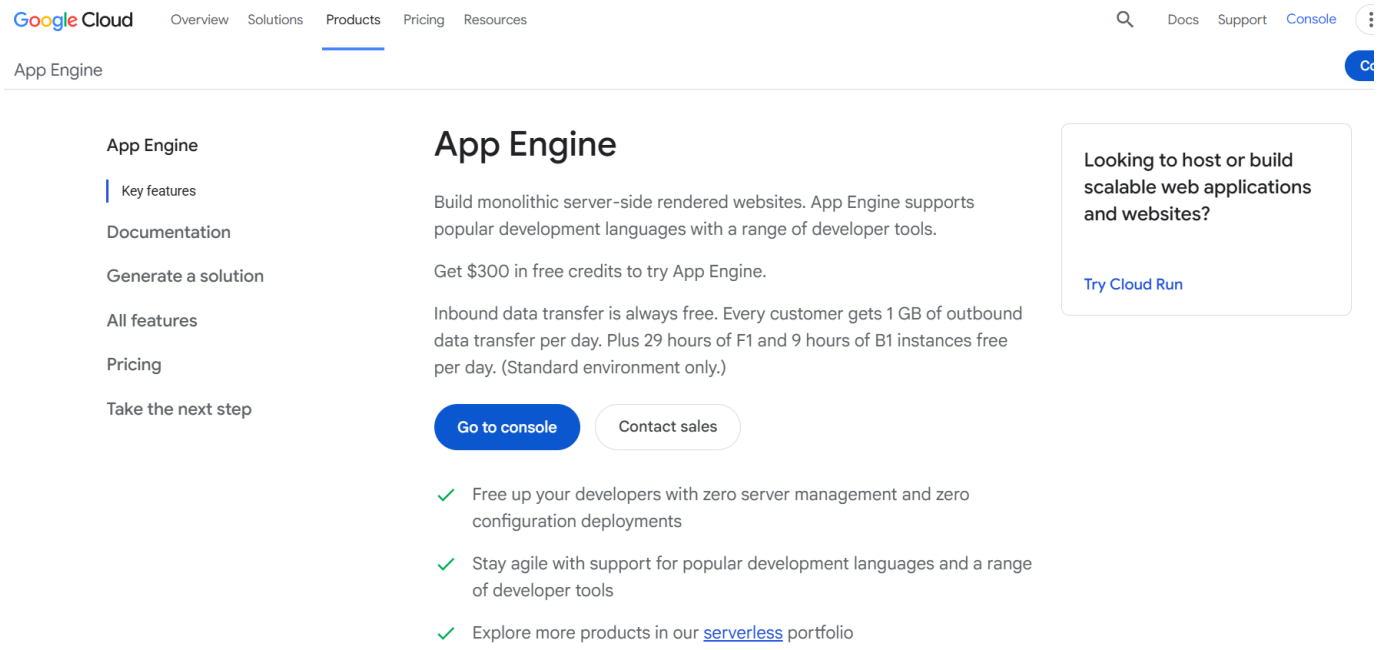
WEEK-7: Use GAE launcher to launch the web applications.

Google App Engine (GAE): A platform-as-a-service (PaaS) offering from Google Cloud that allows developers to build, deploy, and scale applications without managing the underlying infrastructure.

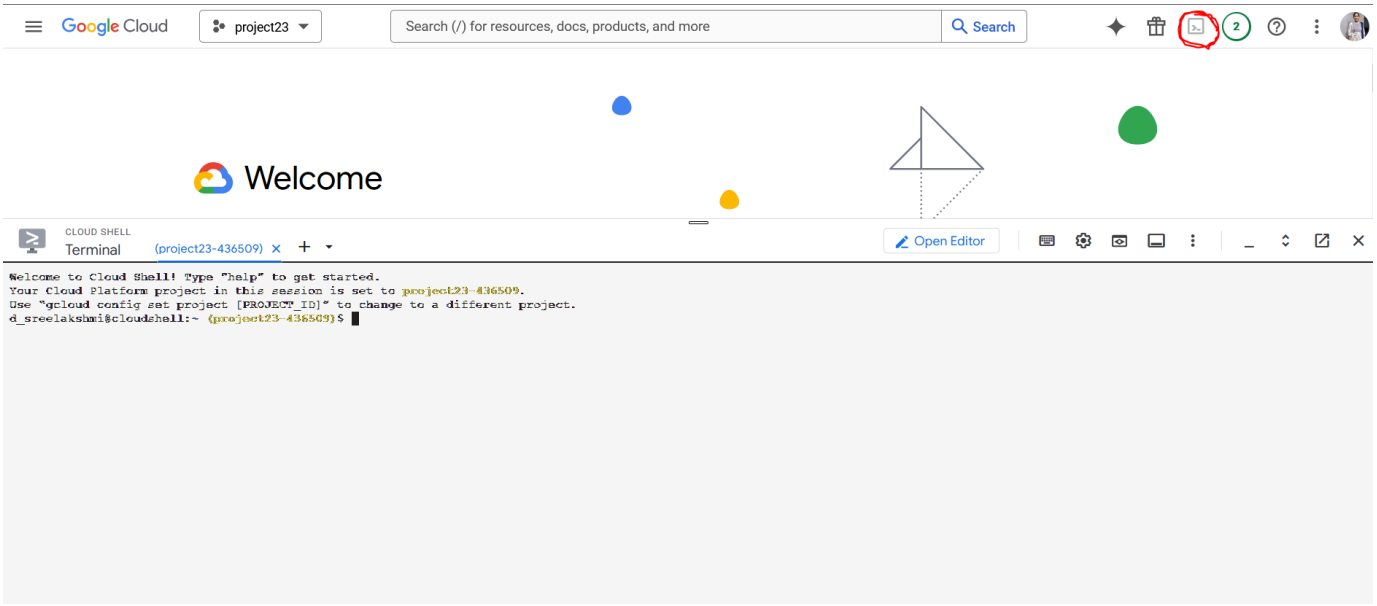
GAE Launcher: Google Cloud Shell is an interactive, browser-based shell environment for managing and interacting with Google Cloud resources. It provides a command-line interface (CLI) that comes preconfigured with essential tools and libraries needed to work with Google Cloud.

1. Signin to Google App Engine: click on console



2. Open Google Cloud Shell:

- Go to the Google Cloud Console.
- Click the Cloud Shell icon in the top right corner. This will open a terminal in your browser with all the tools pre-installed.



3. Create a New Directory:

- In the Cloud Shell terminal, create a new directory for your web app
  mkdir 24251A1226

- Change to the new directory
  mkdir 24251A1226

4: Set Up Your Google Cloud Project

1. Authenticate with Google Cloud:
   gcloud auth login

2. Create a new project:
   gcloud projects create ccweek71226

3. Verify the list of projects:
   gcloud projects list

4. Set the project ID in the CLI:
   gcloud config set project ccweek71226

5. Describe the project:
   gcloud projects describe ccweek71226

5. Create a Simple Python Web Application:

nano app.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def home():
    return render_template('webapp.html')

if __name__ == '__main__':
    app.run(debug=True)
```

Create the webapp.html in a templates folder

Place your webapp.html in a folder named templates in the same directory as app.py. The directory structure should look like this:

```
/your_project_directory
    /templates
        webapp.html
    app.py
```

nano webapp.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Simple Counter App</title>
    <style>
```

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
  font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica, Arial, sans-serif;
}

body {
  background-color: #f2f4f3;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
}

.counter-card {
  background-color: #ffffff;
  width: 380px;
  padding: 45px 30px;
  border-radius: 12px;
  box-shadow: 0 4px 15px rgba(0, 0, 0, 0.05);
  text-align: center;
}

.title {
  font-size: 2rem;
  font-weight: 700;
  color: #2b2b2b;
  margin-bottom: 25px;
}

.count-display {
  font-size: 4.5rem;
```

```
    font-weight: 700;
    color: #2ecc71;
    margin-bottom: 35px;
}

.button-group {
    display: flex;
    justify-content: center;
    gap: 8px;
}

.btn {
    flex: 1;
    padding: 12px 6px;
    border: none;
    border-radius: 6px;
    color: #ffffff;
    font-weight: 700;
    font-size: 0.95rem;
    cursor: pointer;
    transition: opacity 0.2s ease;
    line-height: 1.2;
}

.btn:hover {
    opacity: 0.9;
}

.btn-decrease {
    background-color: #e74c3c;
}

.btn-reset {
```

```
      background-color: #95a5a6;
    }

    .btn-increase {
      background-color: #2ecc71;
    }
  </style>
</head>
<body>

  <div class="counter-card">
    <h1 class="title">Counter App</h1>
    <div id="count" class="count-display">0</div>
    <div class="button-group">
      <button class="btn btn-decrease" onclick="decrease()">-
<br>Decrease</button>
      <button class="btn btn-reset" onclick="reset()">Reset</button>
      <button class="btn btn-increase"
onclick="increase()">+<br>Increase</button>
    </div>
  </div>

  <script>
    let count = 0;
    const countDisplay = document.getElementById('count');

    function updateDisplay() {
      countDisplay.textContent = count;
    }

    function increase() {
      count++;
      updateDisplay();
    }
```

```
        function decrease() {
            count--;
            updateDisplay();
        }

        function reset() {
            count = 0;
            updateDisplay();
        }
    </script>

</body>
</html>
```

4. Create the App Engine Configuration File

nano app.yaml

```
runtime: python39

handlers:
 - url: /.*
   script: auto
```

This configuration tells Google App Engine to use Python 3.9 and run the application.

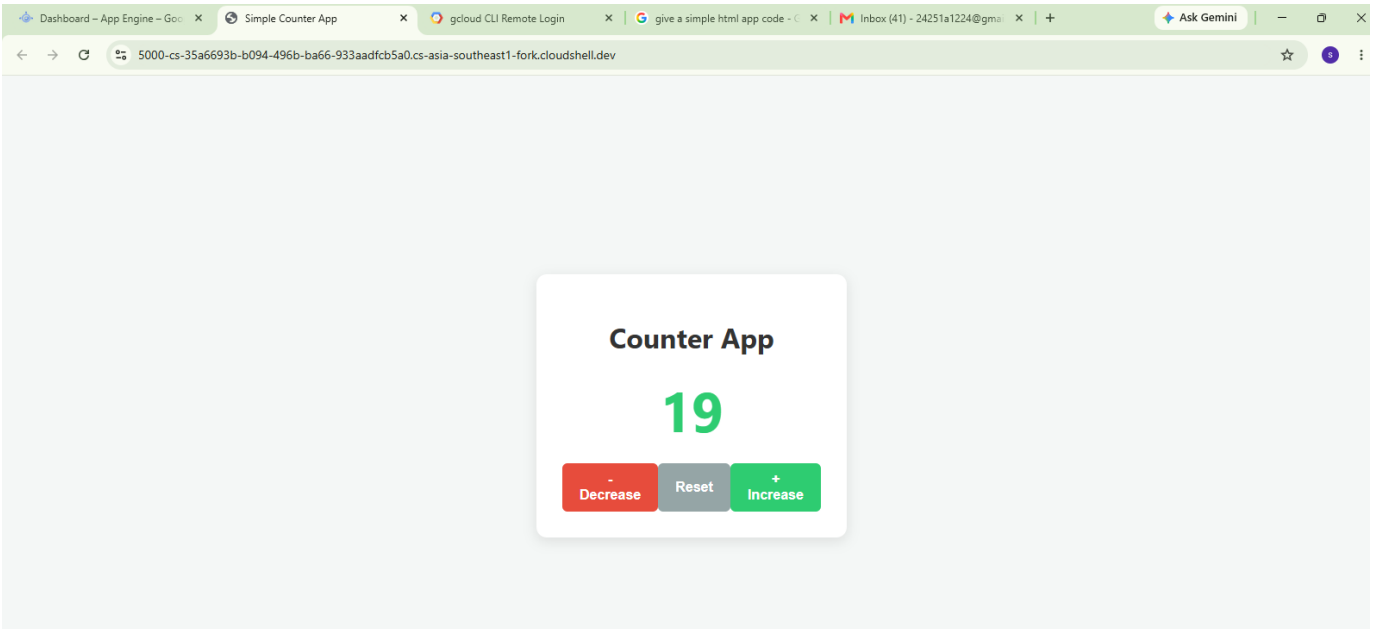
5. Deploy the Application

gcloud app deploy

7. To View your application

gcloud app browse

python app.py



OUTPUT: Welcome to Cloud Shell! Type "help" to get started.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
g24251a1226@cloudshell:~$ gcloud auth login

You are already authenticated with gcloud when running
inside the Cloud Shell and so do not need to run this
command. Do you wish to proceed anyway?

Do you want to continue (Y/n)? Y

Go to the following link in your browser, and complete the sign-in prompts:

https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Fsdk.cloud.google.com%2Fauthcode.html&scope=openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fsqlservice.login+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth&state=nfj4zzZqrUPyRnKAOmntDcV5N8QQqC&prompt=consent&token_usage=remote&access_type=offline&code_challenge=tD-GmYygYIUmhvU23I2TI_6TLb0braCe0gQOikt587I&code_challenge_method=S256

Once finished, enter the verification code provided in your browser:
4/0ATsMZqBhHNUSLhAhpBTz98WY-mSDhQnjJL2lAODWOfCZlmr8CnmZhHoptmca6wOVpxso4Q

```
You are now logged in as  24251a1226@gmail.com .
Your current project is [None].  You can change this setting by running:
  $ gcloud config set project PROJECT_ID
g24251a1226@cloudshell:~$ gcloud projects list
PROJECT_ID: cc-week-6-1226
NAME: cc-week-6-1226
PROJECT_NUMBER: 696311775722
ENVIRONMENT:

PROJECT_ID: ccexam1226
NAME: ccexam1226
PROJECT_NUMBER: 289308500469
ENVIRONMENT:

PROJECT_ID: gen-lang-client-0577760573
NAME: Default Gemini Project
PROJECT_NUMBER: 996059227593
ENVIRONMENT:

PROJECT_ID: project-5297cb27-aebb-491a-842
NAME: My First Project
PROJECT_NUMBER: 827700224332
ENVIRONMENT:

PROJECT_ID: sinuous-radar-cq6d2
NAME:
PROJECT_NUMBER: 739087448918
ENVIRONMENT:
g24251a1226@cloudshell:~$ gcloud projects create ccweek7
ERROR: (gcloud.projects.create) Project creation failed. The project ID you specified is already in use by
another project. Please try an alternative ID.
g24251a1226@cloudshell:~$ gcloud projects create ccweek71226
Create in progress for  https://cloudresourcemanager.googleapis.com/v1/projects/ccweek71226 .
Waiting for [operations/create_project.global.8306578818124998524] to finish...done.
Enabling service [cloudapis.googleapis.com] on project [ccweek71226]...
Operation "operations/acat.p2-184891232526-ea02f44f-9c04-4274-b7d6-aca2d51b7d54" finished
successfully.
g24251a1226@cloudshell:~$ gcloud config set project ccweek71226
Updated property [core/project].
g24251a1226@cloudshell:~ (ccweek71226)$ nano app.py
g24251a1226@cloudshell:~ (ccweek71226)$ nano requirements.txt
g24251a1226@cloudshell:~ (ccweek71226)$ nano app.yaml
g24251a1226@cloudshell:~ (ccweek71226)$ mkdir templates
g24251a1226@cloudshell:~ (ccweek71226)$ cd templates
g24251a1226@cloudshell:~/templates (ccweek71226)$ nano webapp.html
g24251a1226@cloudshell:~/templates (ccweek71226)$ cd ..
g24251a1226@cloudshell:~ (ccweek71226)$ nano app.py
g24251a1226@cloudshell:~ (ccweek71226)$ python app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production
WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with watchdog (inotify)
 * Debugger is active!
 * Debugger PIN: 118-217-953
```

```
127.0.0.1 - - [09/Sep/2026 09:11:57] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [09/Sep/2026 09:11:57] "GET /favicon.ico HTTP/1.1" 404 -
^Z
[1]+  Stopped                 python app.py
g24251a1226@cloudshell:~ (ccweek71226)$ nano app.py
g24251a1226@cloudshell:~ (ccweek71226)$ nano app.yaml
g24251a1226@cloudshell:~ (ccweek71226)$ nano requirements.txt
g24251a1226@cloudshell:~ (ccweek71226)$ cd templates
g24251a1226@cloudshell:~/templates (ccweek71226)$ nano webapp.html
g24251a1226@cloudshell:~/templates (ccweek71226)$ cd ..
g24251a1226@cloudshell:~ (ccweek71226)$ gcloud app deploy
WARNING: You might be using automatic scaling for a standard environment deployment, without
providing a value for automatic_scaling.max_instances. Starting from March, 2025, App Engine sets the
automatic scaling maximum instances default for standard environment deployments to 20. This change
doesn't impact existing apps. To override the default, specify the new max_instances value in your
app.yaml file, and deploy a new version or redeploy over an existing version. For details on
max_instances, see https://cloud.google.com/appengine/docs/standard/reference/app-
yaml.md#scaling_elements.

You are creating an app for project [ccweek71226].
WARNING: Creating an App Engine application for a project is irreversible and the region
cannot be changed. More information about regions is at
<https://cloud.google.com/appengine/docs/locations>.

WARNING: Starting from March, 2025, App Engine sets the automatic scaling maximum instances
default for standard environment deployments to 20. This change doesn't impact
existing apps. To override the default, specify the new max_instances value in your
app.yaml file, and deploy a new version or redeploy over an existing version.
For more details on max_instances, see
<https://cloud.google.com/appengine/docs/standard/reference/app-yaml.md#scaling_elements>.

NOTE: Cloud Run offers the most modern fully managed application hosting experience
with lower minimum billable times and support for GPUs on demand for your AI/ML workloads.
Deploy code written in any programming language supported by App Engine on Cloud Run.
Learn more at https://cloud.google.com/run/docs/quickstarts#build-and-deploy-a-web-service

Please choose the region where you want your App Engine application located:

 [1] asia-east1    (supports standard and flexible)
 [2] asia-east2    (supports standard and flexible and search_api)
 [3] asia-northeast1 (supports standard and flexible and search_api)
 [4] asia-northeast2 (supports standard and flexible and search_api)
 [5] asia-northeast3 (supports standard and flexible and search_api)
 [6] asia-south1   (supports standard and flexible and search_api)
 [7] asia-southeast1 (supports standard and flexible)
 [8] asia-southeast2 (supports standard and flexible and search_api)
 [9] australia-southeast1 (supports standard and flexible and search_api)
 [10] europe-central2 (supports standard and flexible)
 [11] europe-west   (supports standard and flexible and search_api)
 [12] europe-west2  (supports standard and flexible and search_api)
 [13] europe-west3  (supports standard and flexible and search_api)
 [14] europe-west6  (supports standard and flexible and search_api)
 [15] northamerica-northeast1 (supports standard and flexible and search_api)
 [16] southamerica-east1 (supports standard and flexible and search_api)
 [17] us-central    (supports standard and flexible and search_api)
 [18] us-east1      (supports standard and flexible and search_api)
 [19] us-east4      (supports standard and flexible and search_api)
```

```
 [20] us-west1      (supports standard and flexible)
 [21] us-west2      (supports standard and flexible and search_api)
 [22] us-west3      (supports standard and flexible and search_api)
 [23] us-west4      (supports standard and flexible and search_api)
 [24] cancel
Please enter your numeric choice:  1

ERROR: (gcloud.app.deploy) INVALID_ARGUMENT: The project must have a billing account attached.
g24251a1226@cloudshell:~ (ccweek71226)$
```

WEEK-8

Simulate a cloud scenario using CloudSim and run a scheduling algorithm in CloudSim

Simulation

- Simulation is making a virtual model of a real system to see how it would act in different situations.
- It helps us test and understand systems safely without changing the real thing.

CloudSim

- CloudSim is an open-source framework, which is used to simulate cloud computing infrastructure and services.
- It is developed by the CLOUDS Lab organization and is written entirely in Java.
- It is used for modelling and simulating a cloud computing environment.

Some of the most common classes used during simulation are:

- **Datacenter:** used for modelling the foundational hardware equipment of any cloud environment, that is the Datacenter.
- **Host:** this class executes actions related to management of virtual machines. It also defines policies for provisioning memory and bandwidth to the virtual machines, as well as allocating CPU cores to the virtual machines.
- **VM:** this class represents a virtual machine by providing data members defining a VM's bandwidth, RAM, mips (million instructions per second), size while also providing setter and getter methods for these parameters.
- **Cloudlet:** a cloudlet class represents any task that is run on a VM, like a processing task, or a memory access task, or a file updating task etc.
- **CloudSim:** this is the class responsible for initializing and starting the simulation environment after all the necessary cloud entities have been defined, and later stopping after all the entities have been destroyed.

Benefits of CloudSim

- Open source and free of cost
- Easy to use and scalable
- Risks can be evaluated at an earlier stage
- Provides pre-defined simulation packages

Scheduling Algorithm

In CloudSim, a scheduling algorithm schedules how cloud resources, like virtual machines (VMs), are assigned to different tasks or jobs.

Steps

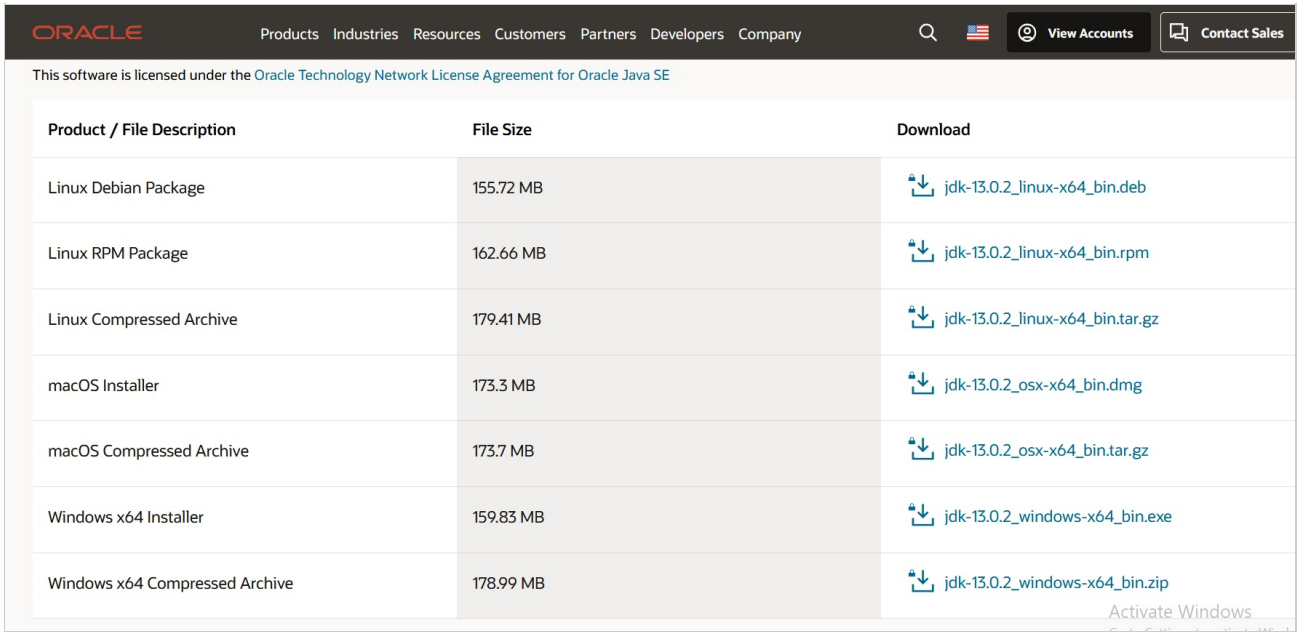
1. Java Installation

a. Check Java in your system

Open Command Prompt (Windows) or Terminal (Mac/Linux). Type the following command to check the Java version:

```
java -version
```

If Java is installed, you'll see the version details. If not installed, proceed to download and install Java from the official Java download page.



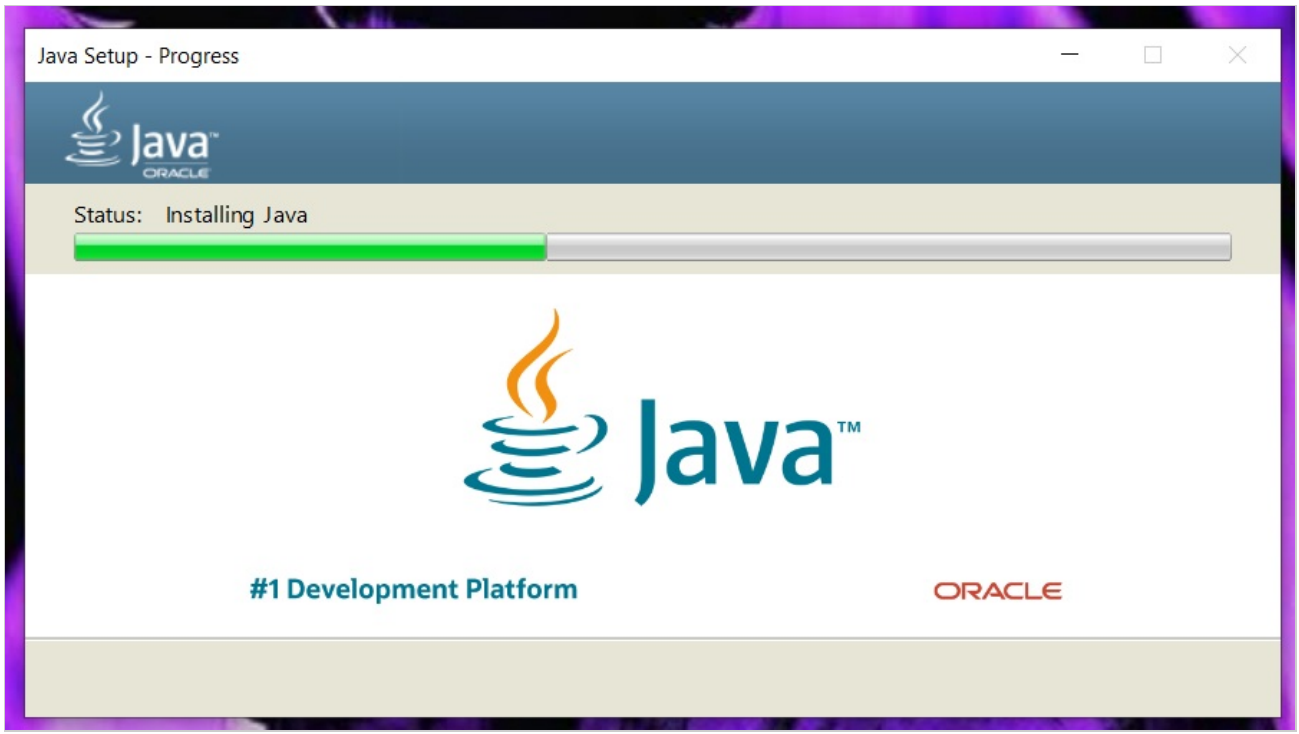
Java SE downloads page

https://www.oracle.com/java/technologies/javase/jdk13-archive-downloads.html

b. Download the appropriate version for your operating system (64-bit)

- For Windows, download the **.exe** file.
- For Mac, download the **.dmg** file.
- For Linux, download the **.tar.gz** file.

c. Install Java — run the downloaded setup, follow the installation instructions, and complete the installation process.



Java installation wizard

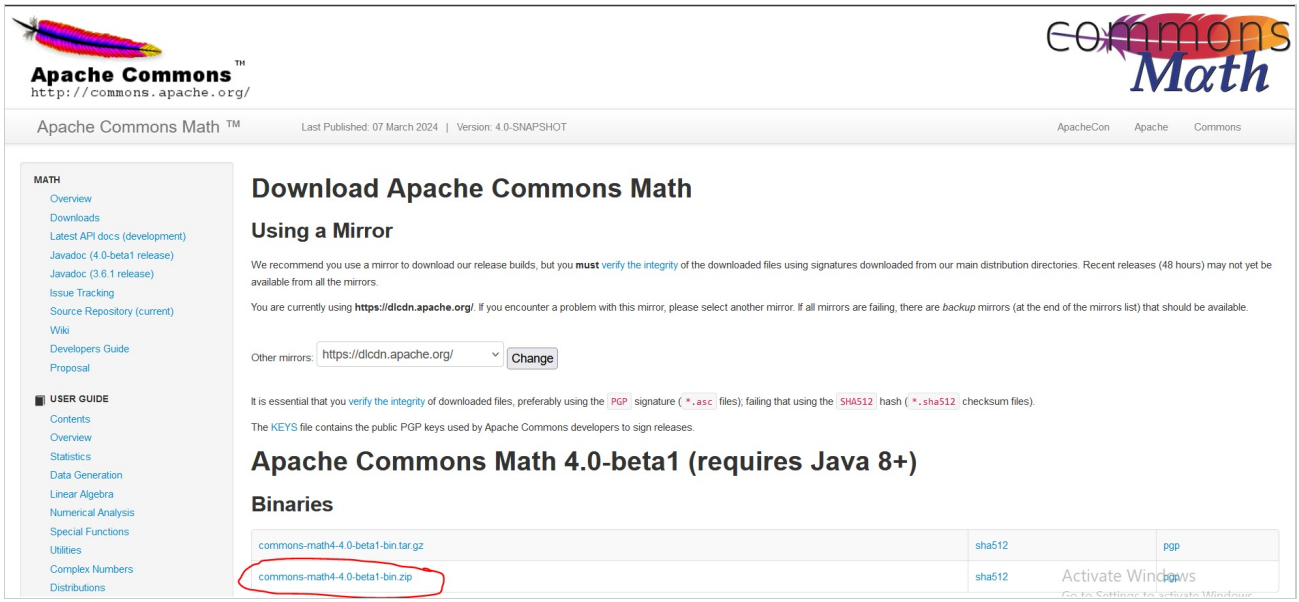
d. Set Path Variable for Java (Windows)

- Right-click on This PC or My Computer and go to Properties.
- Select Advanced system settings.
- Click Environment Variables.
- In System variables, find Path, select it, and click Edit.
- Add the path of the bin folder in the Java installation directory: **C:\Program Files\Java\jdk-14\bin**
- Click OK to save the changes.

2. Download CloudSim and additional jar file

a. Download CloudSim — download CloudSim and extract the contents.

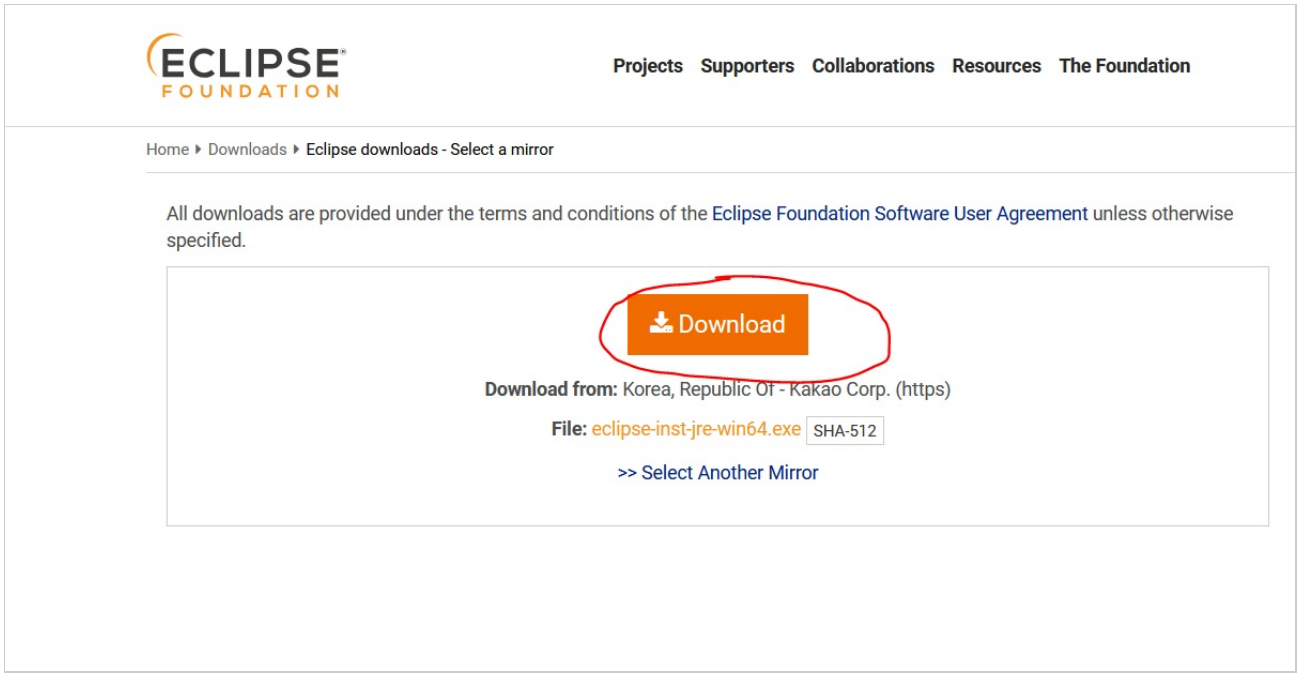
b. Download Commons Math 3 jar file



Commons Math 3 jar download

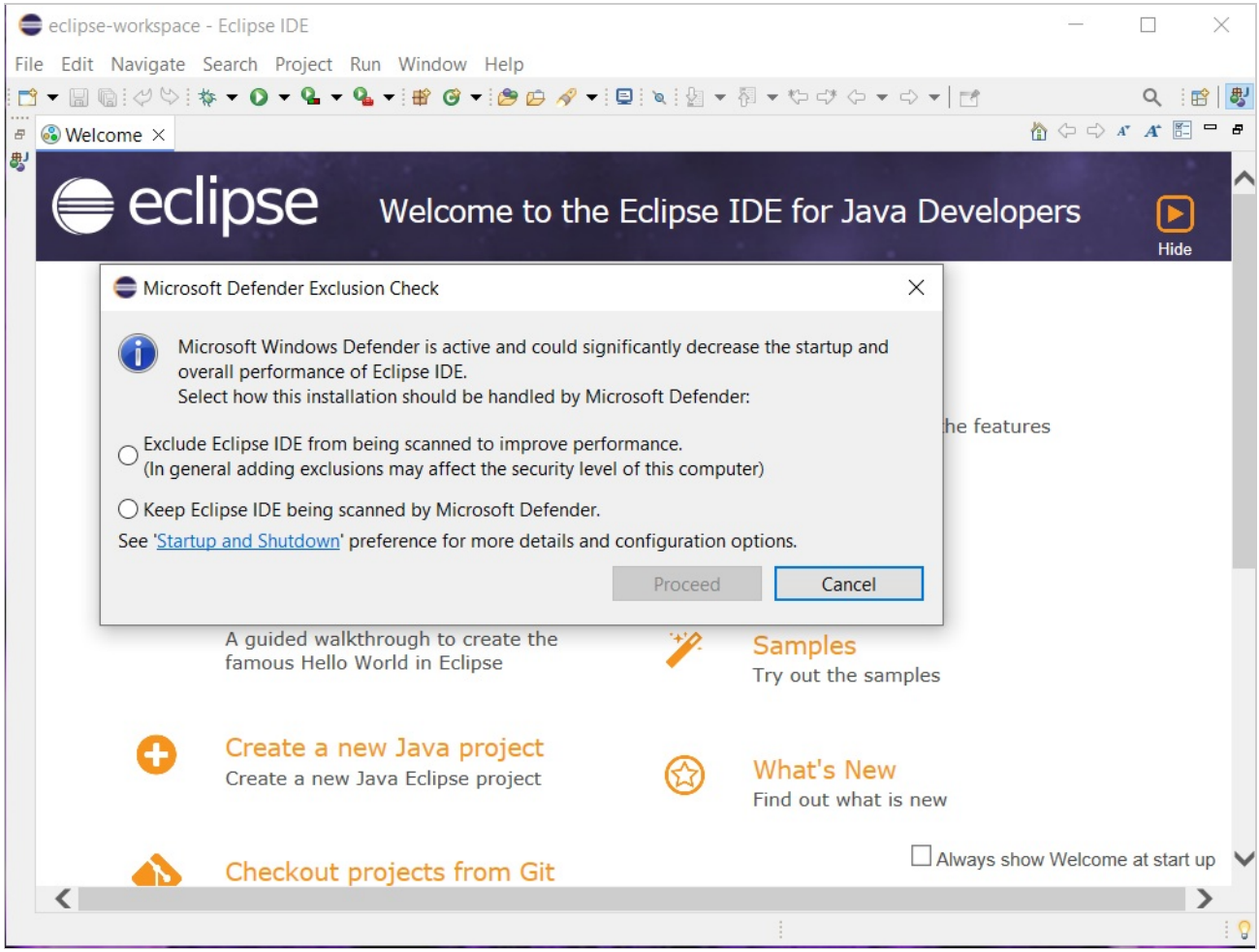
3. Eclipse IDE Installation

- If Java 64-bit is installed, then download 64-bit Eclipse.



Eclipse IDE download

- Install Eclipse IDE.



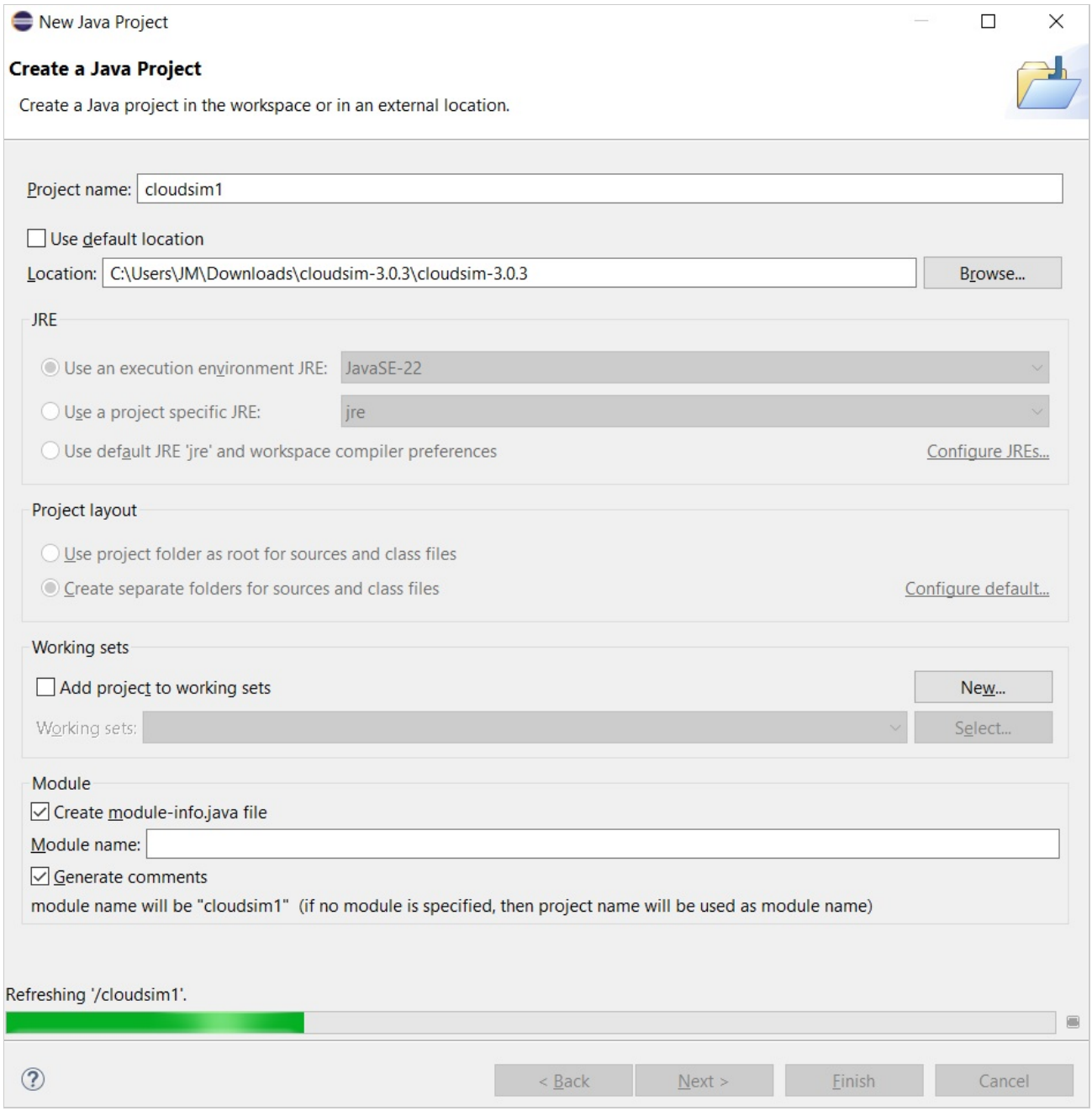
Eclipse IDE installer

4. Run CloudSim in Eclipse

- Put the commons-math jar file into the jar folder of CloudSim 3.0.3.

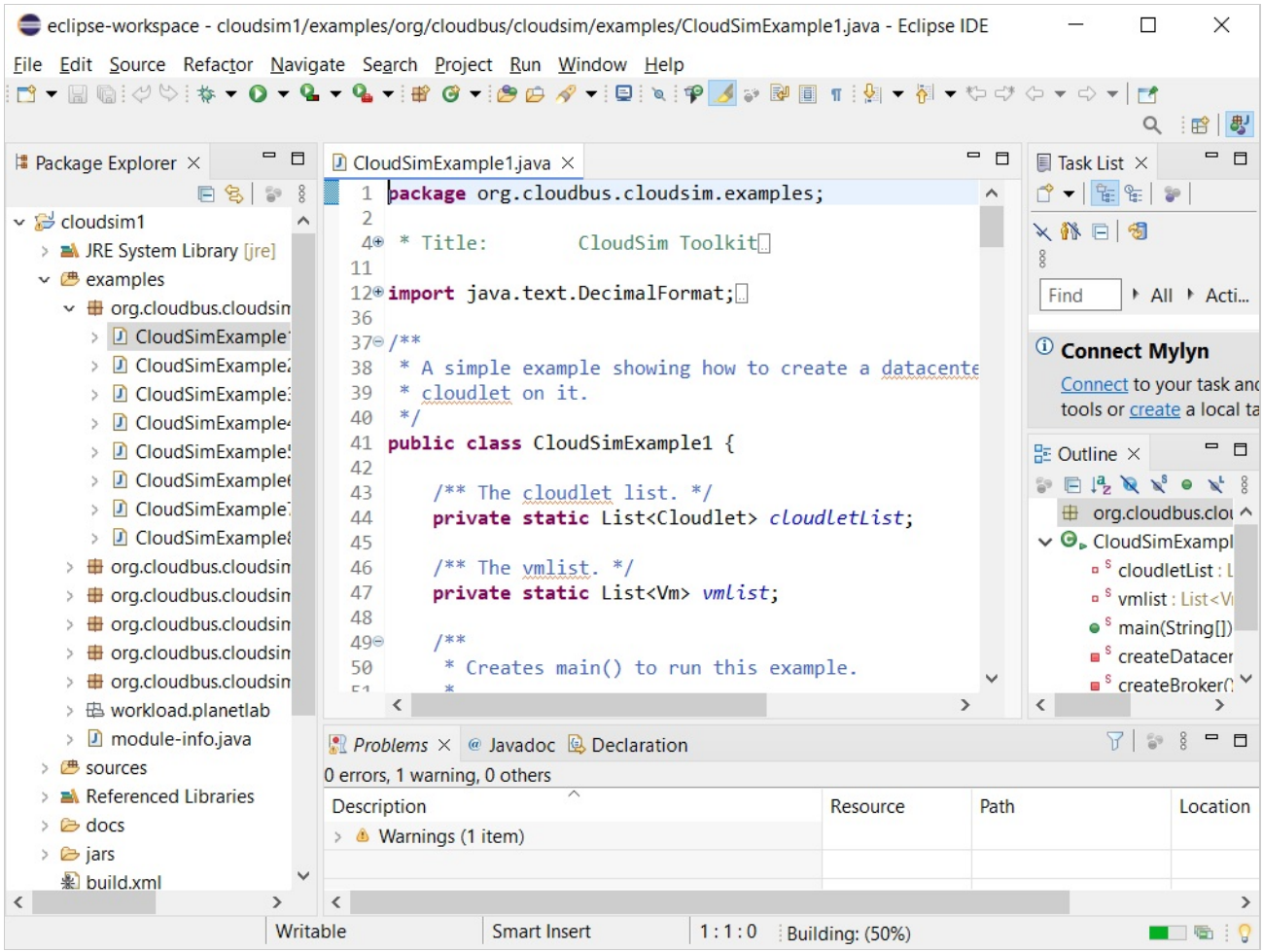
- Build a new Java project with the CloudSim 3.0.3 folder.

Open Eclipse and create a new Java project.



Creating a new Java project in Eclipse

After creation of the project, execute any example from the CloudSim examples.



Running a CloudSim example in Eclipse

Example CloudSim Program: Creating a Datacenter, VM and Simulation

SimpleVmCreation.java

```
package org.cloudbus.cloudsim.examples;

import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.Vm;

public class SimpleVmCreation {

    public static void main(String[] args) {
        try {
            // Initialize CloudSim
            CloudSim.init(1, null, false);

            // Create a VM
            Vm vm = createVm();

            // Print VM information
            System.out.println("VM created: " + vm.getId());

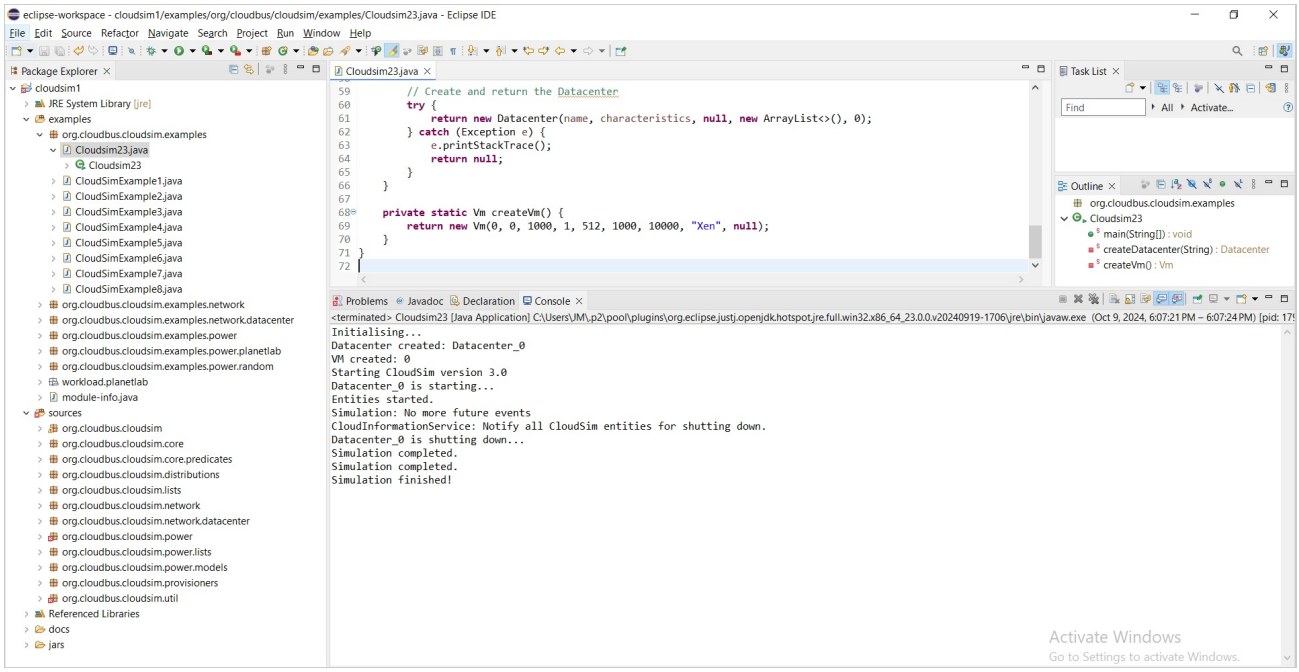
            // Start the simulation (although no datacenter is created)
            CloudSim.startSimulation();
            CloudSim.stopSimulation();

            System.out.println("Simulation finished!");
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    private static Vm createVm() {
        // Create and return a VM with specified parameters
        return new Vm(0, 0, 1000, 1, 512, 1000, 10000, "Xen", null);
    }
}
```

VM Parameters

Parameter	Value / Description
ID	Unique identifier for the VM
Number of Cores	Set to 1
MIPS (Million Instructions Per Second)	1000 (processing power)
RAM	512 MB
Bandwidth	1000 Mbps
Storage	10,000 MB
Hypervisor	"Xen" (type of virtualization)



Console output of the SimpleVmCreation example

Program and Output

TestExample.java

```
package org.cloudbus.cloudsim.examples.network.datacenter;

import java.io.IOException;
import java.text.DecimalFormat;
import java.util.ArrayList;
import java.util.Calendar;
import java.util.LinkedList;
import java.util.List;

import org.cloudbus.cloudsim.Cloudlet;
import org.cloudbus.cloudsim.DatacenterCharacteristics;
import org.cloudbus.cloudsim.Host;
import org.cloudbus.cloudsim.Log;
import org.cloudbus.cloudsim.Pe;
import org.cloudbus.cloudsim.Storage;
import org.cloudbus.cloudsim.VmSchedulerTimeShared;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.network.datacenter.EdgeSwitch;
import org.cloudbus.cloudsim.network.datacenter.NetDatacenterBroker;
import org.cloudbus.cloudsim.network.datacenter.NetworkConstants;
import org.cloudbus.cloudsim.network.datacenter.NetworkDatacenter;
import org.cloudbus.cloudsim.network.datacenter.NetworkHost;
import org.cloudbus.cloudsim.network.datacenter.NetworkVm;
import org.cloudbus.cloudsim.network.datacenter.NetworkVmAllocationPolicy;
import org.cloudbus.cloudsim.provisioners.BwProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.PeProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.RamProvisionerSimple;

public class TestExample {

    /** The vmlist. */
    private static List<NetworkVm> vmlist;

    /**
     * Creates main() to run this example.
     *
     * @param args the args
     */
    public static void main(String[] args) {

        Log.printLine("Starting CloudSimExample1...");

        try {
            int num_user = 1; // number of cloud users
            Calendar calendar = Calendar.getInstance();
            boolean trace_flag = false; // mean trace events

            // Initialize the CloudSim library
            CloudSim.init(num_user, calendar, trace_flag);

            // Second step: Create Datacenters
            // Datacenters are the resource providers in CloudSim. We need at
            // list one of them to run a CloudSim simulation
            NetworkDatacenter datacenter0 = createDatacenter("Datacenter_0");

            // Third step: Create Broker
            NetDatacenterBroker broker = createBroker();
            broker.setLinkDC(datacenter0);

            // Fifth step: Create one Cloudlet
            vmlist = new ArrayList<NetworkVm>();

            // submit vm list to the broker
            broker.submitVmList(vmlist);

            // Sixth step: Starts the simulation
            CloudSim.startSimulation();
            CloudSim.stopSimulation();

            // Final step: Print results when simulation is over
            List<Cloudlet> newList = broker.getCloudletReceivedList();
            printCloudletList(newList);
            System.out.println("numberofcloudlet " + newList.size() + " Cached "
                    + NetDatacenterBroker.cachedcloudlet + " Data transfered "
                    + NetworkConstants.totaldatatransfer);

            Log.printLine("CloudSimExample1 finished!");
        } catch (Exception e) {
            e.printStackTrace();
            Log.printLine("Unwanted errors happen");
        }
    }
```

```
    /**
     * Creates the datacenter.
     *
     * @param name the name
     * @return the datacenter
     */
    private static NetworkDatacenter createDatacenter(String name) {

        List<NetworkHost> hostList = new ArrayList<NetworkHost>();

        int mips = 1;

        int ram = 2048; // host memory (MB)
        long storage = 1000000; // host storage
        int bw = 10000;
        for (int i = 0; i < NetworkConstants.EdgeSwitchPort * NetworkConstants.AggSwitchPort
                * NetworkConstants.RootSwitchPort; i++) {
            List<Pe> peList = new ArrayList<Pe>();
            peList.add(new Pe(0, new PeProvisionerSimple(mips)));
            peList.add(new Pe(1, new PeProvisionerSimple(mips)));
            peList.add(new Pe(2, new PeProvisionerSimple(mips)));
            peList.add(new Pe(3, new PeProvisionerSimple(mips)));
            peList.add(new Pe(4, new PeProvisionerSimple(mips)));
            peList.add(new Pe(5, new PeProvisionerSimple(mips)));
            peList.add(new Pe(6, new PeProvisionerSimple(mips)));
            peList.add(new Pe(7, new PeProvisionerSimple(mips)));

            hostList.add(new NetworkHost(
                    i,
                    new RamProvisionerSimple(ram),
                    new BwProvisionerSimple(bw),
                    storage,
                    peList,
                    new VmSchedulerTimeShared(peList)));
        }

        String arch = "x86";
        String os = "Linux";
        String vmm = "Xen";
        double time_zone = 10.0;
        double cost = 3.0;
        double costPerMem = 0.05;
        double costPerStorage = 0.001;
        double costPerBw = 0.0;
        LinkedList<Storage> storageList = new LinkedList<Storage>();

        DatacenterCharacteristics characteristics = new DatacenterCharacteristics(
                arch, os, vmm, hostList, time_zone, cost, costPerMem, costPerStorage, costPerBw);

        NetworkDatacenter datacenter = null;
        try {
            datacenter = new NetworkDatacenter(
                    name, characteristics, new NetworkVmAllocationPolicy(hostList), storageList, 0);
        } catch (Exception e) {
            e.printStackTrace();
        }
        // Create Internal Datacenter network
        CreateNetwork(2, datacenter);
        return datacenter;
    }

    /**
     * Creates the broker.
     *
     * @return the datacenter broker
     */
    private static NetDatacenterBroker createBroker() {
        NetDatacenterBroker broker = null;
        try {
            broker = new NetDatacenterBroker("Broker");
        } catch (Exception e) {
            e.printStackTrace();
            return null;
        }
        return broker;
    }

    /**
     * Prints the Cloudlet objects.
     *
     * @param list list of Cloudlets
     * @throws IOException
     */
    private static void printCloudletList(List<Cloudlet> list) throws IOException {
        int size = list.size();
        Cloudlet cloudlet;
        String indent = "    ";
        Log.printLine();
        Log.printLine("========== OUTPUT ==========");
        Log.printLine("Cloudlet ID" + indent + "STATUS" + indent + "Data center ID" + indent + "VM ID"
```

```
                + indent + "Time" + indent + "Start Time" + indent + "Finish Time");

        DecimalFormat dft = new DecimalFormat("###.##");
        for (int i = 0; i < size; i++) {
            cloudlet = list.get(i);
            Log.print(indent + cloudlet.getCloudletId() + indent + indent);

            if (cloudlet.getCloudletStatus() == Cloudlet.SUCCESS) {
                Log.print("SUCCESS");
                Log.printLine(indent + indent + cloudlet.getResourceId() + indent + indent + indent
                        + cloudlet.getVmId() + indent + indent + dft.format(cloudlet.getActualCPUTime())
                        + indent + indent + dft.format(cloudlet.getExecStartTime()) + indent + indent
                        + dft.format(cloudlet.getFinishTime()));
            }
        }
    }

    static void CreateNetwork(int numhost, NetworkDatacenter dc) {

        // Edge Switch
        EdgeSwitch edgeswitch[] = new EdgeSwitch[1];

        for (int i = 0; i < 1; i++) {
            edgeswitch[i] = new EdgeSwitch("Edge" + i, NetworkConstants.EDGE_LEVEL, dc);
            dc.Switchlist.put(edgeswitch[i].getId(), edgeswitch[i]);
        }

        for (Host hs : dc.getHostList()) {
            NetworkHost hs1 = (NetworkHost) hs;
            hs1.bandwidth = NetworkConstants.BandWidthEdgeHost;
            int switchnum = (int) (hs.getId() / NetworkConstants.EdgeSwitchPort);
            edgeswitch[switchnum].hostlist.put(hs.getId(), hs1);
            dc.HostToSwitchid.put(hs.getId(), edgeswitch[switchnum].getId());
            hs1.sw = edgeswitch[switchnum];
            List<NetworkHost> hslist = hs1.sw.fintimelistHost.get(0D);
            if (hslist == null) {
                hslist = new ArrayList<NetworkHost>();
                hs1.sw.fintimelistHost.put(0D, hslist);
            }
            hslist.add(hs1);
        }
    }
}
```

Terminal Output

```
Starting CloudSimExample1...
Initialising...
Starting CloudSim version 3.0
Datacenter_0 is starting...
Edge0 is starting...
Broker is starting...
Entities started.
0.0: Broker: Cloud Resource List received with 1 resource(s)
0 VM is created on 0
1 VM is created on 1
2 VM is created on 2
3 VM is created on 3
4 VM is created on 0
5 VM is created on 1
6 VM is created on 2
7 VM is created on 3
app0
app1
app2
app3
app4
app5
app6
app7
app8
app9
app10
app11
app12
app13
app14
app15
app16
app17
app18
app19
app20
app21
app22
app23
app24
```

```
app25
app26
app27
app28
app29
app30
app31
app32
app33
app34
app35
app36
app37
app38
app39
app40
app41
app42
app43
app44
app45
app46
app47
app48
app49
app50
app51
app52
app53
app54
app55
app56
app57
app58
app59
app60
app61
app62
app63
app64
app65
app66
app67
app68
app69
app70
app71
app72
app73
app74
app75
app76
app77
app78
app79
app80
app81
app82
app83
app84
app85
app86
app87
app88
app89
app90
app91
app92
app93
app94
app95
app96
app97
app98
app99
Simulation: No more future events
CloudInformationService: Notify all CloudSim entities for shutting down.
Datacenter_0 is shutting down...
Edge0 is shutting down...
Broker is shutting down...
Simulation completed.
Simulation completed.

========== OUTPUT ==========
Cloudlet ID    STATUS    Data center ID    VM ID    Time    Start Time    Finish Time
    1        SUCCESS        2            0        800        0        800
    4        SUCCESS        2            0        800        0        800
    7        SUCCESS        2            0        800        0        800
    10        SUCCESS        2            0        800        0        800
    0        SUCCESS        2            5        800        0        800
```

```
3       SUCCESS       2          5       800       0       800
6       SUCCESS       2          5       800       0       800
9       SUCCESS       2          5       800       0       800
13       SUCCESS       2          0       801       800       1601
16       SUCCESS       2          0       801       800       1601
19       SUCCESS       2          0       801       800       1601
22       SUCCESS       2          0       801       800       1601
12       SUCCESS       2          5       801       800       1601
15       SUCCESS       2          5       801       800       1601
18       SUCCESS       2          5       801       800       1601
21       SUCCESS       2          5       801       800       1601
2       SUCCESS       2          3       1602       0       1602
5       SUCCESS       2          3       1602       0       1602
8       SUCCESS       2          3       1602       0       1602
11       SUCCESS       2          3       1602       0       1602
25       SUCCESS       2          0       801       1601       2402
28       SUCCESS       2          0       801       1601       2402
31       SUCCESS       2          0       801       1601       2402
34       SUCCESS       2          0       801       1601       2402
24       SUCCESS       2          5       801       1601       2402
27       SUCCESS       2          5       801       1601       2402
30       SUCCESS       2          5       801       1601       2402
33       SUCCESS       2          5       801       1601       2402
14       SUCCESS       2          3       803       1602       2405
17       SUCCESS       2          3       803       1602       2405
20       SUCCESS       2          3       803       1602       2405
23       SUCCESS       2          3       803       1602       2405
37       SUCCESS       2          0       801       2402       3203
40       SUCCESS       2          0       801       2402       3203
43       SUCCESS       2          0       801       2402       3203
46       SUCCESS       2          0       801       2402       3203
36       SUCCESS       2          5       801       2402       3203
39       SUCCESS       2          5       801       2402       3203
42       SUCCESS       2          5       801       2402       3203
45       SUCCESS       2          5       801       2402       3203
26       SUCCESS       2          3       803       2405       3208
29       SUCCESS       2          3       803       2405       3208
32       SUCCESS       2          3       803       2405       3208
35       SUCCESS       2          3       803       2405       3208
49       SUCCESS       2          0       801       3203       4004
52       SUCCESS       2          0       801       3203       4004
55       SUCCESS       2          0       801       3203       4004
58       SUCCESS       2          0       801       3203       4004
48       SUCCESS       2          5       801       3203       4004
51       SUCCESS       2          5       801       3203       4004
54       SUCCESS       2          5       801       3203       4004
57       SUCCESS       2          5       801       3203       4004
38       SUCCESS       2          3       803       3208       4011
41       SUCCESS       2          3       803       3208       4011
44       SUCCESS       2          3       803       3208       4011
47       SUCCESS       2          3       803       3208       4011
61       SUCCESS       2          0       801       4004       4805
64       SUCCESS       2          0       801       4004       4805
67       SUCCESS       2          0       801       4004       4805
70       SUCCESS       2          0       801       4004       4805
60       SUCCESS       2          5       801       4004       4805
63       SUCCESS       2          5       801       4004       4805
66       SUCCESS       2          5       801       4004       4805
69       SUCCESS       2          5       801       4004       4805
50       SUCCESS       2          3       803       4011       4814
53       SUCCESS       2          3       803       4011       4814
56       SUCCESS       2          3       803       4011       4814
59       SUCCESS       2          3       803       4011       4814
73       SUCCESS       2          0       801       4805       5606
76       SUCCESS       2          0       801       4805       5606
79       SUCCESS       2          0       801       4805       5606
82       SUCCESS       2          0       801       4805       5606
72       SUCCESS       2          5       801       4805       5606
75       SUCCESS       2          5       801       4805       5606
78       SUCCESS       2          5       801       4805       5606
81       SUCCESS       2          5       801       4805       5606
62       SUCCESS       2          3       803       4814       5617
65       SUCCESS       2          3       803       4814       5617
68       SUCCESS       2          3       803       4814       5617
71       SUCCESS       2          3       803       4814       5617
85       SUCCESS       2          0       801       5606       6407
88       SUCCESS       2          0       801       5606       6407
91       SUCCESS       2          0       801       5606       6407
94       SUCCESS       2          0       801       5606       6407
84       SUCCESS       2          5       801       5606       6407
87       SUCCESS       2          5       801       5606       6407
90       SUCCESS       2          5       801       5606       6407
93       SUCCESS       2          5       801       5606       6407
74       SUCCESS       2          3       803       5617       6420
77       SUCCESS       2          3       803       5617       6420
80       SUCCESS       2          3       803       5617       6420
83       SUCCESS       2          3       803       5617       6420
97       SUCCESS       2          0       801       6407       7208
100       SUCCESS       2          0       801       6407       7208
103       SUCCESS       2          0       801       6407       7208
```

```
106        SUCCESS        2            0        801        6407        7208
96        SUCCESS        2            5        801        6407        7208
99        SUCCESS        2            5        801        6407        7208
102        SUCCESS        2            5        801        6407        7208
105        SUCCESS        2            5        801        6407        7208
86        SUCCESS        2            3        803        6420        7223
89        SUCCESS        2            3        803        6420        7223
92        SUCCESS        2            3        803        6420        7223
95        SUCCESS        2            3        803        6420        7223
109        SUCCESS        2            0        801        7208        8009
112        SUCCESS        2            0        801        7208        8009
115        SUCCESS        2            0        801        7208        8009
118        SUCCESS        2            0        801        7208        8009
108        SUCCESS        2            5        801        7208        8009
111        SUCCESS        2            5        801        7208        8009
114        SUCCESS        2            5        801        7208        8009
117        SUCCESS        2            5        801        7208        8009
98        SUCCESS        2            3        803        7223        8026
101        SUCCESS        2            3        803        7223        8026
104        SUCCESS        2            3        803        7223        8026
107        SUCCESS        2            3        803        7223        8026
121        SUCCESS        2            0        801        8009        8810
124        SUCCESS        2            0        801        8009        8810
127        SUCCESS        2            0        801        8009        8810
130        SUCCESS        2            0        801        8009        8810
120        SUCCESS        2            5        801        8009        8810
123        SUCCESS        2            5        801        8009        8810
126        SUCCESS        2            5        801        8009        8810
129        SUCCESS        2            5        801        8009        8810
110        SUCCESS        2            3        803        8026        8829
113        SUCCESS        2            3        803        8026        8829
116        SUCCESS        2            3        803        8026        8829
119        SUCCESS        2            3        803        8026        8829
133        SUCCESS        2            0        801        8810        9611
136        SUCCESS        2            0        801        8810        9611
139        SUCCESS        2            0        801        8810        9611
142        SUCCESS        2            0        801        8810        9611
132        SUCCESS        2            5        801        8810        9611
135        SUCCESS        2            5        801        8810        9611
138        SUCCESS        2            5        801        8810        9611
141        SUCCESS        2            5        801        8810        9611
122        SUCCESS        2            3        803        8829        9632
125        SUCCESS        2            3        803        8829        9632
128        SUCCESS        2            3        803        8829        9632
131        SUCCESS        2            3        803        8829        9632
145        SUCCESS        2            0        801        9611        10412
148        SUCCESS        2            0        801        9611        10412
151        SUCCESS        2            0        801        9611        10412
154        SUCCESS        2            0        801        9611        10412
144        SUCCESS        2            5        801        9611        10412
147        SUCCESS        2            5        801        9611        10412
150        SUCCESS        2            5        801        9611        10412
153        SUCCESS        2            5        801        9611        10412
134        SUCCESS        2            3        803        9632        10435
137        SUCCESS        2            3        803        9632        10435
140        SUCCESS        2            3        803        9632        10435
143        SUCCESS        2            3        803        9632        10435
157        SUCCESS        2            0        801        10412        11213
160        SUCCESS        2            0        801        10412        11213
163        SUCCESS        2            0        801        10412        11213
166        SUCCESS        2            0        801        10412        11213
156        SUCCESS        2            5        801        10412        11213
159        SUCCESS        2            5        801        10412        11213
162        SUCCESS        2            5        801        10412        11213
165        SUCCESS        2            5        801        10412        11213
146        SUCCESS        2            3        803        10435        11238
149        SUCCESS        2            3        803        10435        11238
152        SUCCESS        2            3        803        10435        11238
155        SUCCESS        2            3        803        10435        11238
169        SUCCESS        2            0        801        11213        12014
172        SUCCESS        2            0        801        11213        12014
175        SUCCESS        2            0        801        11213        12014
178        SUCCESS        2            0        801        11213        12014
168        SUCCESS        2            5        801        11213        12014
171        SUCCESS        2            5        801        11213        12014
174        SUCCESS        2            5        801        11213        12014
177        SUCCESS        2            5        801        11213        12014
158        SUCCESS        2            3        803        11238        12041
161        SUCCESS        2            3        803        11238        12041
164        SUCCESS        2            3        803        11238        12041
167        SUCCESS        2            3        803        11238        12041
181        SUCCESS        2            0        801        12014        12815
184        SUCCESS        2            0        801        12014        12815
187        SUCCESS        2            0        801        12014        12815
190        SUCCESS        2            0        801        12014        12815
180        SUCCESS        2            5        801        12014        12815
183        SUCCESS        2            5        801        12014        12815
186        SUCCESS        2            5        801        12014        12815
189        SUCCESS        2            5        801        12014        12815
170        SUCCESS        2            3        803        12041        12844
```

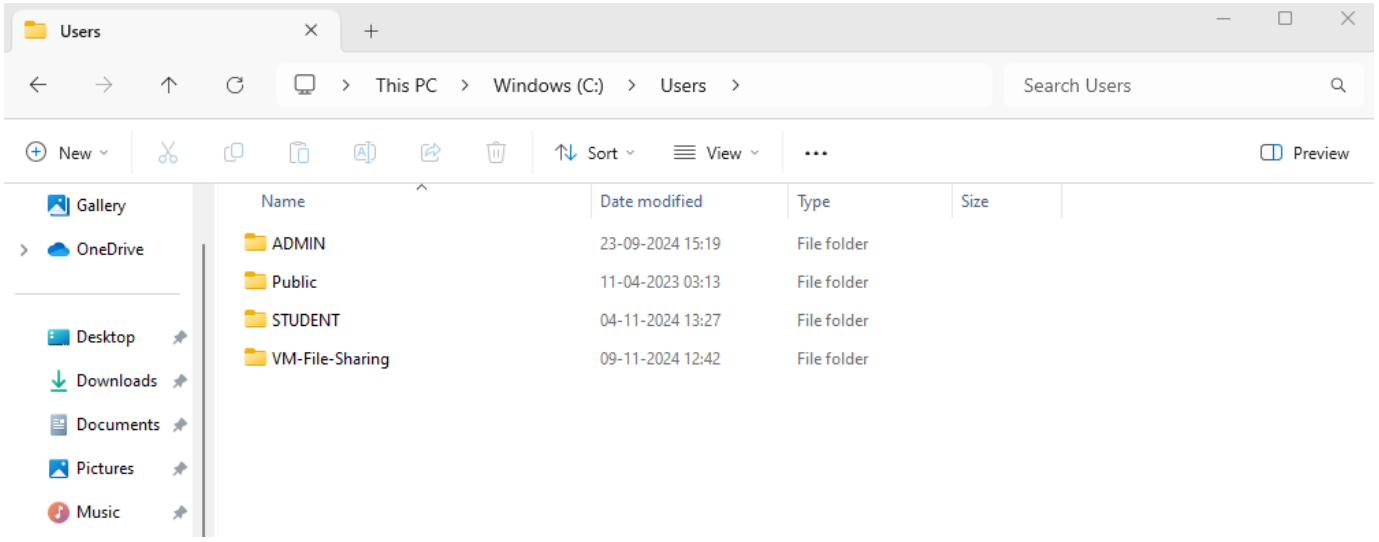
173	SUCCESS	2	3	803	12041	12844
176	SUCCESS	2	3	803	12041	12844
179	SUCCESS	2	3	803	12041	12844
193	SUCCESS	2	0	801	12815	13616
196	SUCCESS	2	0	801	12815	13616
199	SUCCESS	2	0	801	12815	13616
202	SUCCESS	2	0	801	12815	13616
192	SUCCESS	2	5	801	12815	13616
195	SUCCESS	2	5	801	12815	13616
198	SUCCESS	2	5	801	12815	13616
201	SUCCESS	2	5	801	12815	13616
182	SUCCESS	2	3	803	12844	13647
185	SUCCESS	2	3	803	12844	13647
188	SUCCESS	2	3	803	12844	13647
191	SUCCESS	2	3	803	12844	13647
205	SUCCESS	2	0	801	13616	14417
208	SUCCESS	2	0	801	13616	14417
211	SUCCESS	2	0	801	13616	14417
214	SUCCESS	2	0	801	13616	14417
204	SUCCESS	2	5	801	13616	14417
207	SUCCESS	2	5	801	13616	14417
210	SUCCESS	2	5	801	13616	14417
213	SUCCESS	2	5	801	13616	14417
194	SUCCESS	2	3	803	13647	14450
197	SUCCESS	2	3	803	13647	14450
200	SUCCESS	2	3	803	13647	14450
203	SUCCESS	2	3	803	13647	14450
217	SUCCESS	2	0	801	14417	15218
220	SUCCESS	2	0	801	14417	15218
223	SUCCESS	2	0	801	14417	15218
226	SUCCESS	2	0	801	14417	15218
216	SUCCESS	2	5	801	14417	15218
219	SUCCESS	2	5	801	14417	15218
222	SUCCESS	2	5	801	14417	15218
225	SUCCESS	2	5	801	14417	15218
206	SUCCESS	2	3	803	14450	15253
209	SUCCESS	2	3	803	14450	15253
212	SUCCESS	2	3	803	14450	15253
215	SUCCESS	2	3	803	14450	15253
229	SUCCESS	2	0	801	15218	16019
232	SUCCESS	2	0	801	15218	16019
235	SUCCESS	2	0	801	15218	16019
238	SUCCESS	2	0	801	15218	16019
228	SUCCESS	2	5	801	15218	16019
231	SUCCESS	2	5	801	15218	16019
234	SUCCESS	2	5	801	15218	16019
237	SUCCESS	2	5	801	15218	16019
218	SUCCESS	2	3	803	15253	16056
221	SUCCESS	2	3	803	15253	16056
224	SUCCESS	2	3	803	15253	16056
227	SUCCESS	2	3	803	15253	16056
241	SUCCESS	2	0	801	16019	16820
244	SUCCESS	2	0	801	16019	16820
247	SUCCESS	2	0	801	16019	16820
250	SUCCESS	2	0	801	16019	16820
240	SUCCESS	2	5	801	16019	16820
243	SUCCESS	2	5	801	16019	16820
246	SUCCESS	2	5	801	16019	16820
249	SUCCESS	2	5	801	16019	16820
230	SUCCESS	2	3	803	16056	16859
233	SUCCESS	2	3	803	16056	16859
236	SUCCESS	2	3	803	16056	16859
239	SUCCESS	2	3	803	16056	16859
253	SUCCESS	2	0	801	16820	17621
256	SUCCESS	2	0	801	16820	17621
259	SUCCESS	2	0	801	16820	17621
262	SUCCESS	2	0	801	16820	17621
252	SUCCESS	2	5	801	16820	17621
255	SUCCESS	2	5	801	16820	17621
258	SUCCESS	2	5	801	16820	17621
261	SUCCESS	2	5	801	16820	17621
242	SUCCESS	2	3	803	16859	17662
245	SUCCESS	2	3	803	16859	17662
248	SUCCESS	2	3	803	16859	17662
251	SUCCESS	2	3	803	16859	17662
265	SUCCESS	2	0	801	17621	18422
268	SUCCESS	2	0	801	17621	18422
271	SUCCESS	2	0	801	17621	18422
274	SUCCESS	2	0	801	17621	18422
264	SUCCESS	2	5	801	17621	18422
267	SUCCESS	2	5	801	17621	18422
270	SUCCESS	2	5	801	17621	18422
273	SUCCESS	2	5	801	17621	18422
254	SUCCESS	2	3	803	17662	18465
257	SUCCESS	2	3	803	17662	18465
260	SUCCESS	2	3	803	17662	18465
263	SUCCESS	2	3	803	17662	18465
277	SUCCESS	2	0	801	18422	19223
280	SUCCESS	2	0	801	18422	19223
283	SUCCESS	2	0	801	18422	19223

```
    286         SUCCESS         2           0         801         18422         19223
    276         SUCCESS         2           5         801         18422         19223
    279         SUCCESS         2           5         801         18422         19223
    282         SUCCESS         2           5         801         18422         19223
    285         SUCCESS         2           5         801         18422         19223
    266         SUCCESS         2           3         803         18465         19268
    269         SUCCESS         2           3         803         18465         19268
    272         SUCCESS         2           3         803         18465         19268
    275         SUCCESS         2           3         803         18465         19268
    289         SUCCESS         2           0         801         19223         20024
    292         SUCCESS         2           0         801         19223         20024
    295         SUCCESS         2           0         801         19223         20024
    298         SUCCESS         2           0         801         19223         20024
    288         SUCCESS         2           5         801         19223         20024
    291         SUCCESS         2           5         801         19223         20024
    294         SUCCESS         2           5         801         19223         20024
    297         SUCCESS         2           5         801         19223         20024
    278         SUCCESS         2           3         803         19268         20071
    281         SUCCESS         2           3         803         19268         20071
    284         SUCCESS         2           3         803         19268         20071
    287         SUCCESS         2           3         803         19268         20071
    290         SUCCESS         2           3         803         20071         20874
    293         SUCCESS         2           3         803         20071         20874
    296         SUCCESS         2           3         803         20071         20874
    299         SUCCESS         2           3         803         20071         20874
numberofcloudlet 300 Cached 0 Data transfered 200000
CloudSimExample1 finished!
```

WEEK-9: Find a procedure to transfer the files from one virtual machine to another virtual machine.

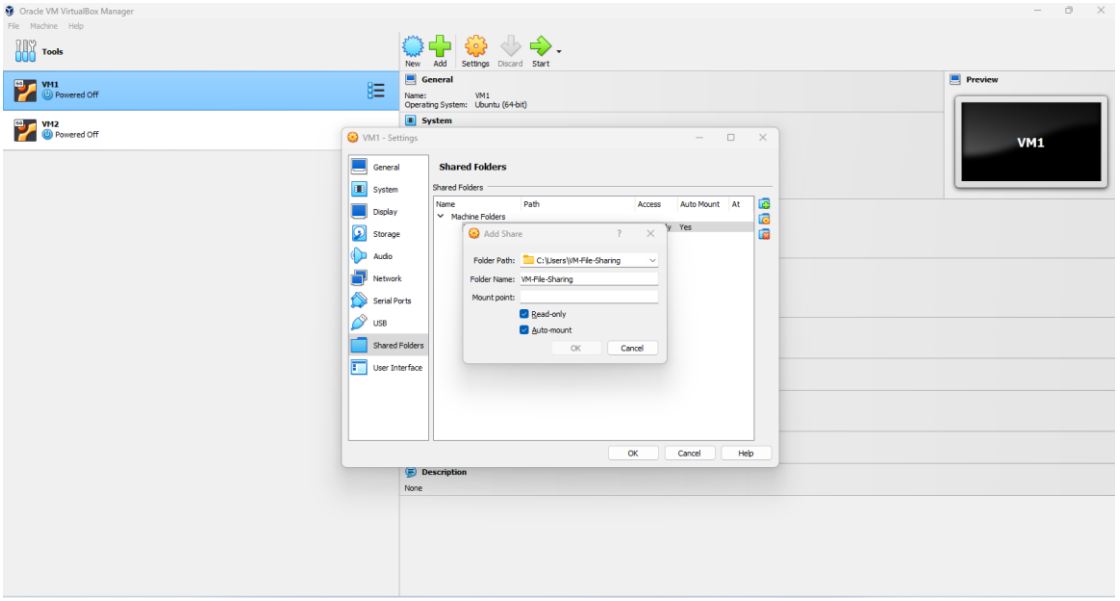
To set up file sharing between your Ubuntu VMs (VM1 and VM2) and the shared folder on your Windows host, follow these steps:

1. First create a Shared folder in C drive



2. Prepare the Shared Folder in VirtualBox Settings:

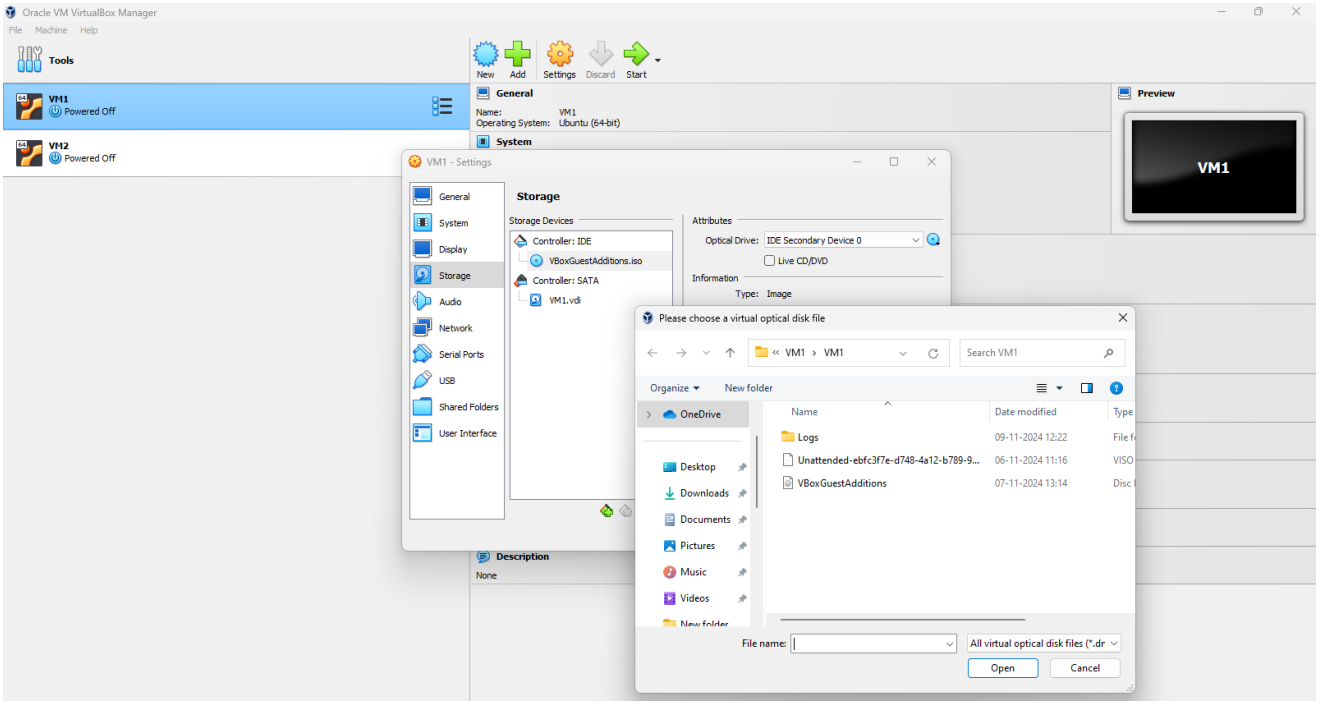
- Shut down VM1 and VM2 if they are running.
- In VirtualBox, select VM1, then go to **Settings > Shared Folders.**
- Click the **+** icon to add a shared folder.
- For **Folder Path**, select **C:\Users\VM-File-Sharing** on your Windows host.
- Set **Folder Name** (e.g., VM-File-Sharing).
- Check both **Auto-mount** and **Make Permanent** to ensure the folder is accessible each time.



- Repeat these steps for VM2, adding the same shared folder.

3. Install Guest Additions in Both VMs:

- Start each VM (VM1 and VM2) one at a time.
- Open VirtualBox, select VM1 / VM2, and go to Settings.
- Under Storage, make sure that the VBoxGuestAdditions.iso is attached as the optical disk.
- In the Storage settings, find Controller: IDE (or similar).
- You should see VBoxGuestAdditions.iso listed under the optical drive. If it says Empty instead, click it, then select Choose a disk file… and locate VBoxGuestAdditions.iso in C:\Users\STUDENT\VirtualBox VMs\VM2\VM2 (or wherever you stored it).



- Once the ISO is selected, click OK to save the changes.Open a terminal in the VM, then enter these commands to install Guest Additions:

  sudo apt update

  sudo apt install build-essential dkms linux-headers-$(uname -r)

  (This command installs necessary packages for compiling and building software (especially kernel modules like the Guest Additions for VirtualBox).

  sudo /media/$USER/VBox_GAs*/VBoxLinuxAdditions.run

  (This command installs the VirtualBox Guest Additions, which are required to improve performance and enable features like file sharing and clipboard integration between the host and the VM.)

- Reboot each VM after installation.

4. Access the Shared Folder in Each VM:

- After rebooting, open a terminal in each VM.
- Add the VM user to the `vboxsf` group to access the shared folder:

  sudo usermod -aG vboxsf $USER

- Log out and log back in for the change to take effect.

5. Locate and Use the Shared Folder:

- To access it, navigate with:
  cd /media/sf_VM-File_sharing
- You can now transfer files by placing them in `C:\Users\VM-File_sharing` on Windows, which will automatically be accessible in the `/media/sf_VM-File_sharing` directory on each VM.

